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Perspectives for HET physics studies at a muon-hadron collider at CERN

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A muon-hadron collider at CERN would extend lepton-proton collisions into the multi-TeV regime and provide a complementary programme of Higgs, electroweak and top (HET) physics. We present the HELP4CERN prospects for 0.5, 1 and 2 TeV muon beams colliding with 7 TeV protons, corresponding to centre-of-mass energies of approximately 3.7, 5.3 and 7.5 TeV. The production benchmarks show substantial gains over the LHeC, particularly for top-quark final states. The accessible deep-inelastic-scattering (DIS) phase space and inclusive production rates provide the basis for the physics programme. Particular attention is given to a dedicated charged-current $h \to b\bar{b}$ study and an exploratory combined Higgs-coupling fit, distinguishing statistical rate precision from coupling constraints that depend on external information and fit assumptions. Electroweak and top measurements, photon-induced production, and muon-coupled new-physics channels offer complementary directions.

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1. Collider benchmarks and HET production rates

The proposed LHμC would collide positive muons with the 7 TeV proton beam of the High-Luminosity LHC, extending the lepton-hadron programme beyond the electron-beam configurations of the LHeC [1–3]. Its higher collision energy opens additional phase space for Higgs, electroweak and top (HET) production, with complementary photon-induced and muon-coupled new-physics channels. Table 1 gives the inclusive production cross-sections for the Higgs and top-quark at the LHμC. The energy dependence is particularly pronounced for heavy final states: relative to the 50 GeV LHeC entries, the 500 GeV and 1 TeV muon configurations increase the single-top cross-section by factors of approximately 7-13 and the top-pair cross-section by 32-69. These comparisons describe the stated production benchmarks; experimental precision additionally depends on luminosity, selection and uncertainties.

Process	20 GeV LHeC	50 GeV LHeC	500 GeV LHμC	1 TeV LHμC
$\sigma(\ell p \to \nu_\ell H + \text{jet} + X)$ [fb]	52	164	438	731
$\sigma(\ell p \to \ell H + \text{jet} + X)$ [fb]	5.6	19	135	221
$\sigma(\ell p \to \ell t\bar{t} + X)$ [pb]	0.0015	0.0323	1.03	2.23
$\sigma(\ell p \to \nu_\ell t + X)$ [pb]	0.55	2.89	21.26	36.77

Table 1: Benchmark inclusive production cross-sections for Higgs-boson, top-quark-pair, and single-top production at the LHeC and LHμC. In the LHeC cases an 80% electron beam polarisation is assumed [6].

The three working points in Table 2 span collision energies from 3.7 to 7.5 TeV. Already the first option enters the multi-TeV regime; the nominal baseline combines increased energy with a larger benchmark sample, while the second option further extends the accessible momentum transfer and final-state masses.

Configuration	E_μ	E_p	$\sqrt{s_{\mu p}}$	$\mathcal{L}_{\text{int}}$
First option	500 GeV	7 TeV	3.7 TeV	250 fb^{-1}
Nominal baseline	1 TeV	7 TeV	5.3 TeV	1000 fb^{-1}
Second option	2 TeV	7 TeV	7.5 TeV	1000 fb^{-1}

Table 2: Benchmark LHμC working points based on the HELP4CERN studies [5].

These working points illustrate the potential of the LHμC physics programme. Already with a 500 GeV muon beam, collisions with 7 TeV protons enter the multi-TeV centre-of-mass regime, providing a basis for competitive studies of Higgs production, electroweak interactions, top-quark processes, photon-induced reactions, and representative scenarios beyond the Standard Model. The nominal baseline combines a higher collision energy with a larger benchmark data sample, supporting more demanding precision measurements and searches. The 2 TeV muon-beam option further extends the accessible momentum transfer and final-state invariant masses. Together, these configurations provide a framework for examining how the physics opportunities develop with collision energy and integrated luminosity.

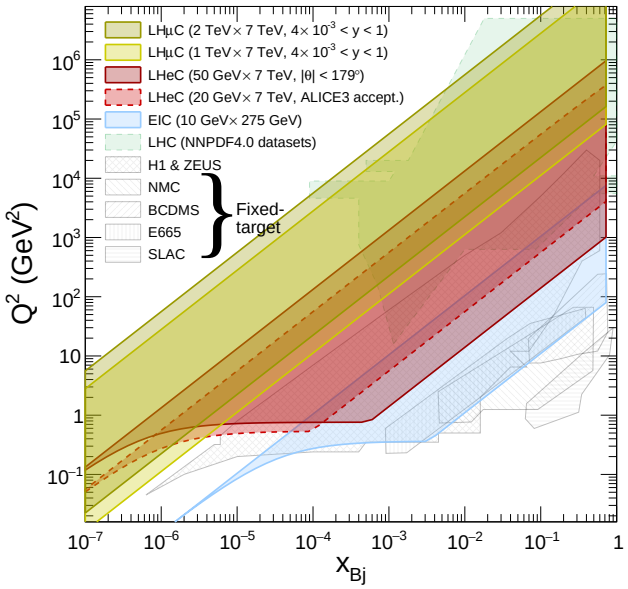


Figure 1: Accessible (x_{Bj}, Q^2) plane from HELP4CERN [5], compared with existing datasets and planned EIC kinematics. The LHμC regions use $4\times10^{-3} < y < 1$. The plotted 20/50 GeV LHeC and 1/2 TeV LHμC configurations retain their original acceptance conventions.

Figure 1 illustrates the enlarged (x_{Bj}, Q^2) coverage. The high-Q^2 reach motivates electroweak tests in deep-inelastic scattering (DIS), alongside the hard-scattering phase space available for HET production. The broader QCD programme also supplies proton-structure information relevant to precision measurements at hadron colliders [3].

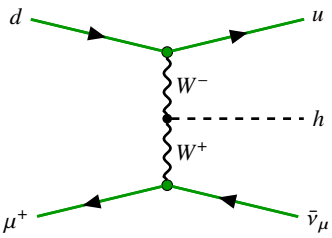
2. Higgs production and precision prospects

The Charged-current (CC) Higgs production proceeds through W^+W^- fusion. The channel $\mu^+p \to j\bar{\nu}_\mu h$ gives an outgoing quark jet, typically forward, and missing momentum from the antineutrino. Neutral-current (NC) production, $\mu^+p \to \mu^+jh$, proceeds through ZZ fusion and retains the scattered charged lepton. Figure 2 shows representative partonic contributions. These complementary signatures provide access to Higgs interactions with weak bosons; extracting couplings also requires decay information and assumptions about the total width.

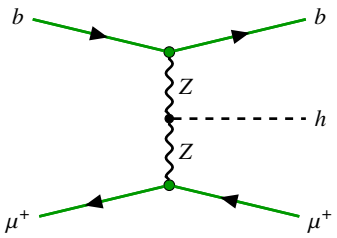
2.1 A dedicated $h \to b\bar{b}$ study

Following the approach of Ref. [7], we study $\mu^+p \to j\bar{\nu}_\mu h$, $h \to b\bar{b}$, against CC multijet production and weak-boson-fusion $Z \to b\bar{b}$. We use an LHeC-like Delphes detector simulation [8], a 75% b-tagging factor, exactly two selected b-jets, and $|m_{bb} - m_h| < 25$ GeV. The statistical rate uncertainty is given by,

$$\frac{\Delta\sigma}{\sigma} = \frac{\sqrt{N_s + N_b}}{N_s}, \tag{1}$$



(a) Charged-current production



(b) Neutral-current production

Figure 2: Representative tree-level diagrams for Higgs production in positive-muon–proton collisions: (a) the charged-current subprocess $\mu^+ d \to \bar{\nu}_\mu u h$ through W^+W^- fusion; (b) the neutral-current subprocess $\mu^+ b \to \mu^+ b h$ through ZZ fusion. The outgoing quark contributes to the jet signature.

where N_s and N_b are selected signal and background counts. The calculated precisions are 1.19% and 0.69% for the 500 GeV and 1 TeV studies, respectively [4, 5].

The bottom-Yukawa extraction uses external information on $R = g^2_{hWW}/\Gamma_h$, where Γ_h is the total Higgs width. Since $\sigma(\mu^+ p \to j \bar{\nu}_\mu h)\,\mathrm{BR}(h \to b\bar{b}) \propto R\, g^2_{hbb}$, independent relative uncertainties are combined as

$$\frac{\Delta g_{hbb}}{g_{hbb}} = \frac{1}{2}\sqrt{\left(\frac{\Delta\sigma}{\sigma}\right)^2 + \left(\frac{\Delta R}{R}\right)^2}. \tag{2}$$

With $\Delta R/R = 0.985\%$ following the FCC-ee benchmark treatment of Ref. [7], the calculated coupling precisions are 0.77% and 0.60%. The analysis thus illustrates subpercent bottom-Yukawa sensitivity, conditional on the external input and statistical treatment, rather than a complete experimental uncertainty forecast.

2.2 Exploratory combined coupling fit

Figure 3 summarizes the complementary κ-3 fit [5, 9], implemented in SMEFiT [10]. The study rescales the statistical components of the LHeC signal-strength uncertainties using production rates while retaining the background, acceptance, efficiency and systematic-factor assumptions.

Relative to the plotted LHeC reference, the combination with the 1 TeV LHμC stage reduces the κ_W, κ_Z and κ_b uncertainties from 0.33%, 0.65% and 1.00% to 0.19%, 0.32% and 0.58%, respectively; the κ_c uncertainty decreases from 3.6% to 1.4%. The combined 500 GeV and 2 TeV stages give κ_W uncertainties of 0.28% and 0.16%, respectively, while the 2 TeV results reach 0.50% for κ_b and 1.2% for κ_c. At 2 TeV, the κ_g and κ_γ uncertainties reach 0.75% and 1.1%. The improvement is channel dependent, with smaller changes in κ_t, κ_μ and $\kappa_{Z\gamma}$. The HL-LHC baselines [11] show the complementary role of these constraints; the "improved" baseline includes reduced proton PDF and α_s uncertainties [3]. Comparisons depend on the width and exotic-decay treatment: the HL-LHC reference study assumes the SM Higgs width and no BSM branching fraction, whereas this fit allows invisible and undetected modes subject to $\kappa_V(\kappa_W, \kappa_Z) \leq 1$. Its vector uncertainty is defined as $1 - \kappa_V(68\%)$. These are exploratory combined projections, distinct from standalone LHμC sensitivities and the dedicated $h \to b\bar{b}$ extraction.

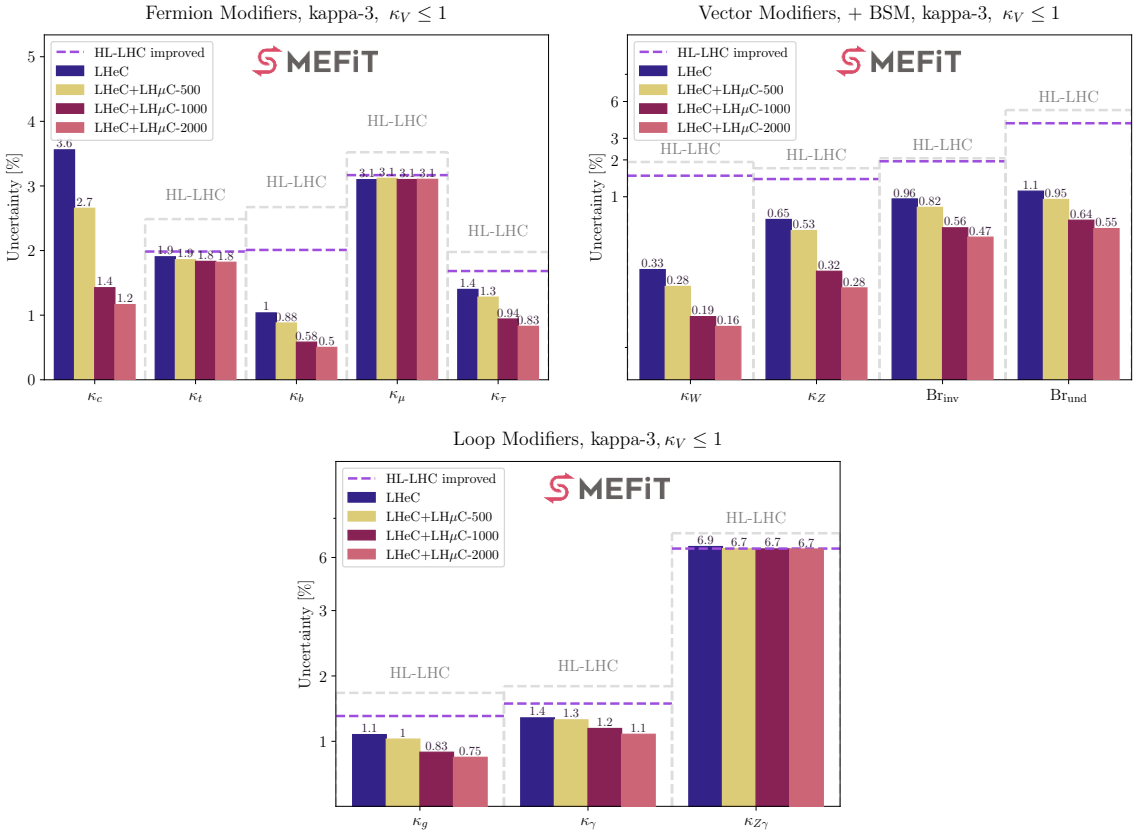


Figure 3: Exploratory κ-3 fit projections from HELP4CERN [5]: fermion modifiers (upper left), vector modifiers and invisible/undetected branching fractions (upper right), and effective loop-induced modifiers (lower). The legend identifies the LHeC and combined LHeC+LHμC stages; dashed lines show HL-LHC baselines. Coupling uncertainties and absolute branching-fraction constraints are in percent. The fit imposes $\kappa_V \leq 1$, with the vector convention defined in the text.

3. Electroweak and top physics

The extended DIS range provides access to electroweak interactions at higher momentum transfer. NC and CC measurements probe different flavour and coupling combinations, motivating studies of the running weak mixing angle, electroweak parameters such as M_W, and effective interactions [2, 5]. The top-production gains in Table 1 make this programme particularly compelling. For the nominal 1 ab^{-1} sample, its cross-sections correspond to approximately 3.7×10^7 produced single-top events and 2.2×10^6 top-pair events, before decays, acceptance and selection. CC single-top production probes the charged-current top interaction and motivates Wtb and CKM studies, while NC top-pair production supplies complementary production and decay observables [5]. The distinct lepton-hadron environment complements the large HL-LHC samples through different initial states and systematic sensitivities, rather than a universal advantage in event counts. Rare decays such as $t \to q\gamma$, qZ and qH, with $q = u, c$, provide further directions for dedicated analyses. Quantitative limits require reconstructed observables, backgrounds and correlated uncertainties. Although low pileup is advantageous, a realistic assessment of detector-level performance requires

detailed detector simulations and comprehensive background studies [5].

4. Photon-induced physics

Quasi-real photons radiated by the lepton and proton provide an additional production mechanism [5, 12, 13]. In the equivalent-photon approximation (EPA) [12], the effective photon-photon luminosity $S_{\gamma\gamma}(W)$ is folded with the subprocess cross section to obtain the total inclusive cross section at the lepton-proton level:

$$\sigma(\mu^+ p \to \mu^+ X p; W > W_0) = \int_{W_0}^{W_{\max}} dW\, S_{\gamma\gamma}(W)\, \hat{\sigma}(\gamma\gamma \to X; W). \tag{3}$$

Here W is the photon-pair invariant mass, distinct from $\sqrt{s_{\mu p}}$, and $W_{\max}$ is its kinematic endpoint. Figure 4 shows the elastic photon-photon luminosity and inclusive τ-pair production rates at the LHeC and LHμC. For $W_0 = 10$ GeV, the reported $\tau^+\tau^-$ cross-sections are 50.37, 58.56 and 70.3 pb for 50 GeV electrons, 500 GeV muons and 1 TeV muons, respectively, on 7 TeV protons [4, 5]. For the same configurations, elastic W^+W^- production above threshold gives 0.037, 0.166 and 0.258 pb. The latter represents enhancements of approximately 4.5 and 7 over the LHeC benchmark, illustrating the stronger benefit for heavier final states.

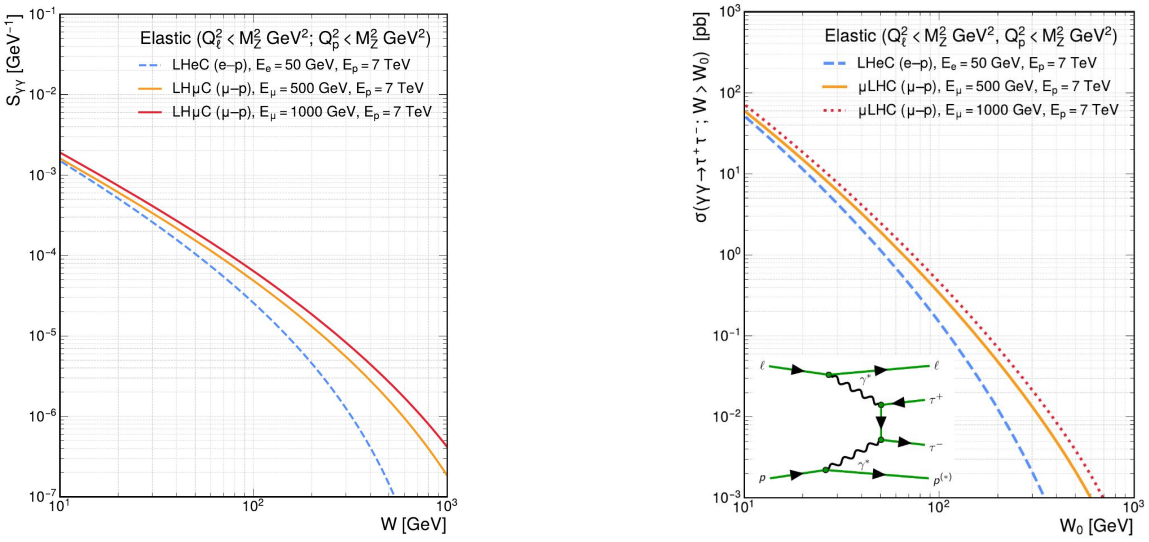


Figure 4: The elastic photon luminosity (left) and cumulative τ-pair cross-section $\sigma(W > W_0)$ (right) at the LHeC and LHμC [4].

The $\gamma\gamma$ programme also includes loop-induced $\gamma\gamma \to ZZ$, top-pair production, and charged BSM pairs such as higgsino-like states and sleptons [4, 5]. The heavier SM channels have smaller, fb-scale rates and require dedicated sensitivity studies. High-mass distributions offer probes of tau electromagnetic properties and gauge-boson interactions.

5. BSM perspectives

Muon-parton interactions provide a further source of complementarity with proton colliders. We consider resonant squark production through R-parity-violating μq interactions and color-octet

muon (leptogluon) production through a μg interaction [5, 7]. In the studied RPV scenarios, the NC channel benefits from a valence-quark initial state relative to the sea-quark CC channel. For color-octet muons, resonant single production at LHμC complements QCD pair production at the HL-LHC. These mechanisms probe the muon coupling directly; their mass reach depends on couplings, the compositeness scale where applicable, luminosity and event selection, rather than on collision energy alone.

6. Conclusions

The LHμC working points extend the CERN lepton-hadron programme to 3.7–7.5 TeV and provide substantial gains in heavy-particle production over the adopted LHeC benchmarks. The dedicated $h \to b\bar{b}$ study illustrates subpercent bottom-Yukawa sensitivity with external coupling/width information, while the exploratory combined fit improves the LHeC vector and fermion constraints, reaching 0.19% for κ_W at the nominal stage under $\kappa_V \leq 1$. Enhanced top production, heavier photon-induced final states and resonant muon-parton channels broaden the physics case. Together with the complementary HL-LHC measurements, these results motivate a coordinated programme of Higgs, electroweak, top and new-physics studies, supported by dedicated detector and systematic analyses.

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